COMSATS University Islamabad
Department of Computer Science
Course Syllabus

Course Information

Course Code: **CSC410** | Course Title: **Professional Practices**
Credit Hours: **3(3,0)** | Lecture Hours/Week: **3**
Lab Hours/Week: **0** | Pre-Requisites: **None**

Catalogue Description:

This course covers the following topics: Introduction to Ethics; Social Implications of Computing & Networked Communications; Intellectual Property; Information Privacy; Privacy & the Government; Computer & Network Security; Computer Reliability; Professional Ethics; and Work & Wealth.

Text and Reference Books

Textbooks:

1. Ethics for the Information Age, Quinn, M.J., 5th Edition, Pearson Education, 2016.
2. Ethical and Social Issues in Information Age, Kizza J. M., 5th Edition, Springer-Verlag, 2017.

Reference Books:

1. Ethics in Information Technology, Reynolds, G., Cengage Course Technology, 2018.
2. A Gift of Fire, Social, Legal, and Ethical Issues for Computing Technology, Baase, S., Pearson Inc, 2017.

Week wise Plan:

Lecture #	CDF Unit #	Topics Covered	Reading Material
1.	1	Introduction to Ethics: Morals & Morality.	Reynolds: Ch1 Quinn: Ch1
2.	1	Ethical Theories: Subjective & Cultural Relativism, Ethical Egoism, and Divine Command Theory.	Reynolds: Ch1
3.	1	Utilitarianism (Act & Rule); and Social Contract Theory.	Reynolds: Ch1
4.	1	Workable Ethical Theories, and Case Study.	Reynolds: Ch1
5.	2	Social Implications of Computing & Networked Communications.	Quinn: Ch3
6.	2	Impact of Social Media on Individualism; and Collectivism & Culture.	Quinn: Ch3
7.	2	Censorship; Freedom of Expression; and Children & Inappropriate Content, and Case Study.	Quinn: Ch3
8.	3	Intellectual Property: Protecting Intellectual, Fair Use, and New Restrictions on Use.	Quinn: Ch4
9.	3	Intellectual Property: Copyrights, Patents, and Trade Marks & Trade Secrets.	Quinn: Ch4
10.	3	Protections for Software; and P2P Networks & Sharing Programs.	Quinn: Ch4
11.	3	Software Licensing; and Open Source Software.	Quinn: Ch4
12.	3	Creative Commons, and Case Study.	Quinn: Ch4
13.	4	Information Privacy; and Perspectives on Privacy.	Quinn: Ch5
14.	4	Information Disclosures.	Quinn: Ch5
15.	4	Data Mining, and Case Study.	Ref. Material
16.	4	Pakistan's Data Protection Act 2020.	DPA2020
17.	**Mid Term Exam**		
18.			

19.	5	Privacy & the Government.	Quinn: Ch6
20.	5	Pakistan's Prevention of Electronic; Crime Bill Act 2016; and Legislation Authorizing Wiretapping.	Quinn: Ch6
21.	5	Regulation of Public & Private Databases; Data Mining by the Government, Information Dissemination, and Case Study.	Quinn: Ch6
22.	6	Ethical Evaluation of Hacking.	Quinn: Ch7
23.	6	Malware; and Cyber Crime & Attacks.	Quinn: Ch7
24.	6	Online Voting, and Case Study.	Quinn: Ch7
25.	7	Computer Reliability; and Data-Entry & Data-Retrieval Errors.	Quinn: Ch8
26.	7	Software & Billing Errors; Notable Software System Failures; and Therac-25.	Quinn: Ch8
27.	7	Computer Simulations; Software Engineering & Software Warranties, and Case Study.	Quinn: Ch8
28.	8	Characteristics of a Fully Developed Profession; and IT Professional vs. Law & Medicine Professionals.	Quinn: Ch9
29.	8	IEEE SE Code of Ethics: Public, Client & Employer, Product, Judgment, and Management.	Quinn: Ch9
30.	8	IEEE SE Code of Ethics: Profession, and Colleagues & Self Related Issues.	Quinn: Ch9
31.	9	Work & Wealth: Automation & Unemployment.	Quinn: Ch10
32.	9	Work Place Changes; Globalization, The Digital Divide, and Case Study.	Quinn: Ch10
Final Term Exam			

Student Outcomes (SOs)

S.#	Description
2	Identify, formulate, research literature, and solve *complex* computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines
7	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice
8	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice
9	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional

Course Learning Outcomes (CLO)

Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy Learning Level	SO
CLO-1	1	Examine the different theories of ethical decision making.	*Understanding*	7-9
CLO-2	2-3	Discuss intellectual property rights for open source software.	Understanding	7-9
CLO-3	4-5	Investigate the impact of technological solutions to privacy issues in the changing world.	*Analyzing*	2,7-9
CLO-4	1-9	Identify ethical issues by examining various case	*Analyzing*	2,7-9

		studies from ICT.		

CLO Assessment Mechanism

Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4
Quizzes	Quiz 1	Quiz 2	Quiz 3	Quiz 4
Assignments	Assignment 1	Assignment 2	Assignment 3	Assignment 4
Mid Term Exam	Mid Term Exam	Mid Term Exam	Mid Term Exam	-
Final Term Exam	Final Term Exam			

Policy & Procedures

- **Attendance Policy:** Every student must attend 80% of the lectures as well as laboratory in this course. The students falling short of required percentage of attendance of lectures/laboratory work, is not allowed to appear in the terminal examination.

- **Course Assessment:**

	Quizzes	Assignments	Mid Term Exam	Terminal Exam	Final Marks
Theory (T)	15	10	25	50	100

- **Grading Policy:** The minimum passing marks for each course is 50% (In case of LAB; in addition to theory, student is also required to obtain 50% marks in the lab to pass the course). The correspondence between letter grades, credit points, and percentage marks at CUI is as follows:

Grade	A	A-	B+	B	B-	C+	C	C-	D+	D	F
Marks	>= 85	80 - 84	75 - 79	71 - 74	68 - 70	64 - 67	61 - 63	58 - 60	54 - 57	50-53	< 50
Cr. Point	3.67-4.00	3.34-3.66	3.01-3.33	2.67-3.00	2.34-2.66	2.01-2.33	1.67-2.00	1.31-1.66	1.01-1.30	0.10-1.00	0.00

- **Missing Exam:** No makeup exam will be given for final exam under any circumstance. When a student misses the mid-term exam for a legitimate reason (such as medical emergencies), his grade for this exam will be determined based on the Department policy. Further, the student must provide an official excuse within one week of the missed exam.

- **Academic Integrity:** All CUI policies regarding ethics apply to this course. The students are advised to discuss their grievances/problems with their counsellors or course instructor in a respectful manner.

- **Plagiarism Policy:** Plagiarism, copying and any other dishonest behaviour is prohibited by the rules and regulations of CUI. Violators will face serious consequences.